EXPLORING INTIMATE SPACE: IN THE PURSUIT OF EMBODIED LISTENING VIA MICRO-SPATIAL SOUND SYSTEMS

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ABSTRACT

This paper explores the development and possibilities for a series of custom-built micro-spatial sound systems suited for composition, performance, software design and installation. Each speaker system (*Speaker_Forest, (solo*), and *(duo)*) is built with hexaphonic and or dodecaphonic audio capabilities, with the potential to be expanded further. In addition, these sound systems are built on a budget and explore alternative modes of sound diffusion through multi-channel tracking in space. The second half of this paper assesses possible spatial composition techniques born from these speaker systems with a focus on embodiment as a method of audience engagement. The findings of this paper reflect the personal experiences of the first author; to this end, the paper makes use of a first-person narrative format in some sections in order to most directly convey the experiential elements of this work.

1. INTRODUCTION

Headphones and personalised listening environments in the home have revolutionised listening since the inception of this technology in the 20th century. Shared listening environments are often politicised in their public spaces by the social context of any given time. These listening environments are likewise often gendered to prioritise the masculine listening experience over others, as noted by sonic arts theorist Holly Ingleton [1]. As someone who has brushed against this hierarchy constantly in my own experience, I became very interested in creating listening spaces where these relationships are put on equal ground. In essence, I pursued the idea of a 'safe' listening space for people, which could also engage acoustic architecture through custom-designed and built sound systems. Diffusion performance was an element to be considered too; the key with this project was to create an open-ended system that would benefit not just performance, but could be used as a tool for composing, for installation and as a tool for new exploring methodologies in sound diffusion.

I am keenly interested in the concept of embodied listening. Katherine Hayles' *How We Became Post-Human* [2] states that we are all subject to cultural constructs, individual experiences, place, timing and spaces (locations, as well as mental space). We know that our relationships to technology become subverted through embodiment: for example, we treat our phones, not as physical objects, but as a transmitter of our loved ones, of work and of our lives. The voice transmitted through these radio waves inhabits a quasi-Anthropomorphic quality to the object itself.

As an artist wanting to conduct people towards phenomenological experiences, it was important to not only develop music, but create opportunities for shared listening 'performances' too. Installations seemed the best opportunity to do this; in their construction they could be primarily concerned with discreet, small-scale spatial sounds whose intimacy encourages listenership between one to three people. The rooms in which these sound systems were set up would also need to be reflective or reverberant to emphasise aspects of the sounds.



Figure 1: The custom-made *Speaker_Forest* sound system and composition tool.

Portable and cost effective spatial installations are few and far between and often the specifics of an installation will stay specific to a piece, rendering their use for one project only. Spatial music technology is a relatively new field of music and is usually privileged to the gallery or university institution. Ultimately the goal to create discreet spatialised works which are portable, cost-effective and could be used in multiple creative iterations by those without university training was the goal to open up the field of spatial works outside of the institution. Inspiring people to engage in a style of sound art they may not have had the opportunity to in the past was one of the core reasons for pursuing embodiment as a primary compositional focus.

Through my phenomenological viewpoint, I aim to explore tools or sound structures in which engage in embodiment and see how music can play a part in generating unique experiences - collective or singular - through custom-designed and built sound sculptures. With this in mind, I set off to create a series of micro-

spatial multi-channel sound installations with the purpose of being able to compose detail-oriented spatial works.



Figure 2: *(solo)* presented as an installation and the first iteration of the speaker dome series

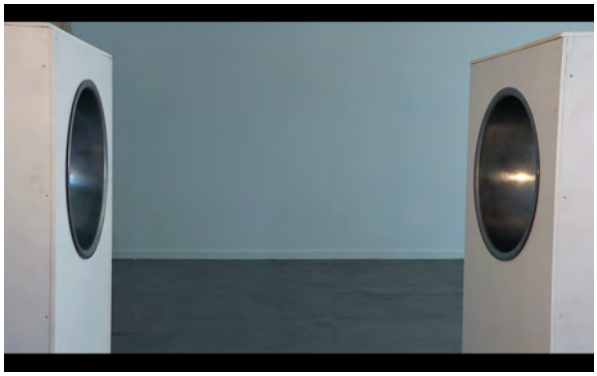


Figure 3: *(duo)*, the second iteration of the speaker dome series.

2. RELATED WORKS

The collection of sound installations which make up the exhibition 'Composing Environments 2.0' (which featured *Speaker_Forest* [figure 1], *(solo)* [figure 2] and *(duo)* [figure 3]) drew from four primary sub-disciplines, pertaining to the development of micro-multichannel sound interfaces, their role in diffusion performances, as well as sculpture and performance art.

2.1. Design and Diffusion

Spatial composition and diffusion performance in electro-acoustic music is a well-established practice, and has diversified hugely in the last 100 years to explore new musical territories. Recently, it even been adopted within the commercial film industry; in 2012 Pixar's *Brave* was the first film to premier the immersive overhead Dolby Atmos system using the 7.1.4 speaker configuration.[1] [2]

Towards the realisation of the *Speaker_Forest* [as seen in figure 1], *(solo)* [figure 2] and *(duo)* [figure 3], there were a number of works that that inspired me to explore these ideas further. Pieces such as Janet Cardiff and George Bures Miller's *40 Part Motet* [figure 4] [3] have been instrumental in the popularization of electroacoustic spatial sound work, and are viewed as occupying sound art and sculptural fields. When the music plays, the empty space becomes activated, unlocking its otherwise forgotten acoustic properties. The speakers are grouped in SATB choral formation singing Thomas Tallis' 1573 *Spem in alium*. The idea of having multiple voices individually represented each by a speaker and their how they engage with the acoustic architecture in the *40 Part Motet* was a key influence in the design of *Speaker_Forest*.



Figure 4: Janet Cardiff and George Bures Miller's *40 Part Motet,* photo courtesy of cardiffmiller.com[3]

Tristan Perich's *Microtonal Wall* [figure 5] differs as it was created with 1500 piezo speakers with dedicated microprocessors which are 'tuned individually, creating a pitch continuum'.[4] Although the scale of this installation showcases a wide tonal range as its sonic output, very finely tuned, it is not able to produce audio signal aside from the tone in which each speaker was initially designed for. The tension begins as you move along the space and listen to the minutia of the speakers as well as the macro-sound picture.



Figure 5: *Microtonal Wall*

Robin Minard's *Silent Music* [figure 6] finds some middle ground between the 40 Part Motet and the Microtonal Wall: it was constructed using minimal materials but using the gallery space, speakers wire and outputting four channel audio. [5] The quadrophonic playback benefits composers through a familiar spatial

1 http://www.hollywoodreporter.com/news/peter-jackson-considering-dolby-atmos-318708

2 http://www.dolby.com/us/en/technologies/home/dolby-atmos.html

3 http://www.cardiffmiller.com/artworks/inst/motet.html#

format. Although the installation is modular the relationship between the speakers is more focussed on the delicacy of the sound world than exploring compositional possibilities with the number of speakers. It focusses instead on the relationship between each of the speakers and how the audio itself interacts with the space, each speaker placement clustering in different configurations depending on the space.

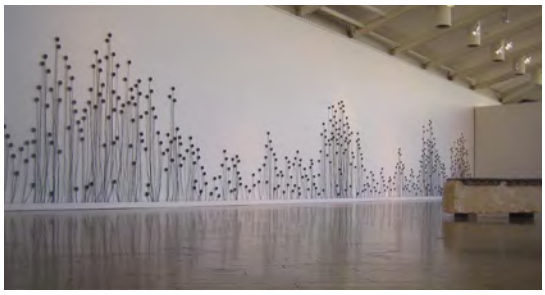


Figure 6: Robin Minard's *Silent Music* courtesy of http://www.robinminard.com/minard.content.php?id=57&sh=0

In undertaking building my work, I wanted to explore the space between each of these works. I wanted to pursue modular installation design that also reflected a wider channel output. This could then be used for finite and delicate composition between each of the smaller speakers. For example in Minard's work there are a maximum of four channels, limiting one's options to compose in a fully spatial way – you can control groups of speakers, but not send audio to the individual speakers themselves, pinpointing it. This focus on speaker arrays became a key focus in the conceptualisation and development of *Speaker_Forest*, *(solo)* and *(duo)*.

2.3. Sculpture and Performance Art

Visual and spatial considerations were important when developing the housing for *(solo)* and *(duo)*; space, reflection and resonance became key elements for the possibility of multiple compositional ideas.

Anish Kapoor's 2008 *Untitled (*figure 7) stainless steel sculpture is one of many iterations in a series by the artist exploring reflection, light and space. [6] Additionally, the curves of the stainless steel hemisphere offer unusual sound reflections, uncovering a unique sound world. The design for the two installations primarily focussed on the type of metal, the depth of the curvature and the shape itself.



Figure 7: Anish Kapoor's *Untitled*, courtesy of: http://anishkapoor.com/721/untitled-21

The layout of elements of *(duo)* and the panels of *Speaker_Forest* draws on the relationship between object and audience, or in this case, performer and reception. In order to experience the works properly, listeners must engage with the work physically, in part due to their position in space and relation to one another. The idea of performed sound sculpture was inspired by the 2010 performance series at MoMA by Marina Abramovic (shown in figure 8).[4] The configuration of shared listening and private listening were main themes in *(solo)* and *(duo)*. The idea of confronting the listening space between two people was especially explored in *(duo)* in the sense that the configuration of the speakers were positioned across from one another making dual listening possible and recommended the way people interact with the space.



Figure 8: Marina Abramovic, *The Artist Is Present* https://www.moma.org/calendar/exhibitions/964?locale=en

3. DESIGN, DEVELOPMENT AND PARTICIPANT EXPERIENCE

This section will discuss the hardware, design and the significance of the design implementation towards the creation of sound work.

3.1 Hardware

Speaker_Forest, *(solo)*, and *(duo)* were all constructed as multi-channel speaker systems with the possibility of being adapted, extended and customised to create unique and dynamic micro-spatial sound experiences for listeners. This section will lay out the design, fabrication, and implementation of these systems. The fabrication of *(solo)* and *(duo)* came in several stages: first, implementing the steel hemispherical design by contracting a metal spinner to make custom-made cases with lips which could be mounted. Next, the construction and mounting of the speakers and amps. Four speakers per amplifier in series-parallel distributes the signal flow

4 https://www.moma.org/calendar/exhibitions/964

evenly to avoid overheating. [5] contains further information about this circuit.

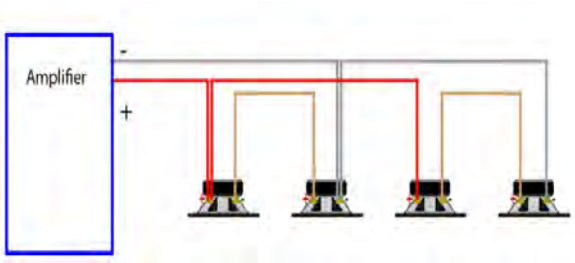


Figure 9: Signal flow from series-parallel circuit.

Each installation is comprised of between six to twelve circuits, and with four speakers per amplifier. The composer can emphasise space and spatial relationships between the speakers through compositional techniques such as panning and phasing, thus 'choreographing' where the sound may travel in the space (a nod to Wishart's notion of spatial motion). How each of the speakers are mounted in their installation also creates delicate and unique signals, some of which move in and out of phase with one another. The best example of this is with *Speaker_Forest,* discussed below.

3.2. Speaker_Forest

Speaker_Forest (shown in Figures 1 and 10) is the first of the micro multi-channel sound installations: it is also a performance tool and diffusion environment for composers and performers. Comprised of twelve audio channels, composers have the opportunity to go beyond usual quadrophonic spatial formats and try hexaphony and dodecaphony. The twelve audio channels are connected through an aggregate audio device and are addressed by Ableton Live. The expansion of multiple channels to be selected over one or more audio interfaces suggests that this installation has the potential to be continually expanded in size and scope to fit a wide range of compositional contexts. The speaker sizes range between 27mm - 40mm creating light-weight, directional structures that float from overhead, forming multi-layered, canopy-like relationships between one another. The audio channel mapping can likewise be customised to the composer's needs as well as the configuration of the panels to one another, which means the possibilities for multiple rigs are up to the mindset of the creator and the pattern of the audio channels as important to consider as sections of an orchestra.

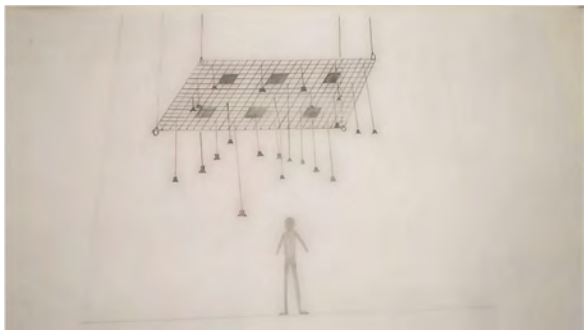


Figure 10: early conceptual sketch of *Speaker_Forest.*

Speaker_Forest hangs above audience members, with the speakers loosely falling down at varying levels, giving listeners the opportunity to physically interact with the space by walking under and through the speakers, like moving through the 'canopies'. The acoustic reflections and arbitrary nature of the way in which the speakers each hang down engage the surrounding space inwardly to be experienced as you walk through them, and outwardly depending on which direction the speakers are facing and what they are aiming. As one walks through the 'canopies', the phasing between each of the speakers creates delicate, timbral textures specific to the height of each listener.

Further documentation of *Speaker_Forest* may be viewed at:
https://www.youtube.com/watch?v=DXZodfD7Ttc.

3.3. *(solo)* & *(duo)*

(solo) (shown in Figure 2, above) is the second iteration in the series of micro-multichannel spatial sound works; a six channel semi-enclosed metal hemisphere is home to twenty-four miniature speakers. This work was constructed with the intention of being primarily used as an installation. The 12mm speakers line the outside of the spheres facing inward, mounted behind the bowl pushing a clear audio signal through 10mm holes. In this work is that the participant can choose to engage by simply leaning into the enclosure. The installation is also battery powered, allowing for maximum portability for outdoor settings. The focus for this piece is on the audio signal and the reflections in the small space itself. The direction of the speakers all facing inwards, and lining the inside of a metal hemisphere gives distorted and immersive reflections, and timbral variation depending on what musical materials are being used.

One can see documentation of *(solo)* at the following link:
https://www.youtube.com/watch?v=q34ewCQ_-D0

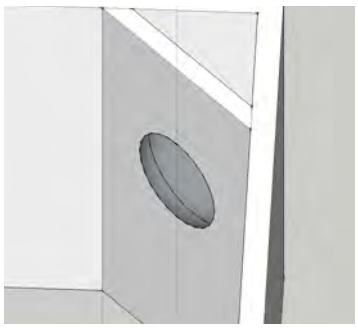


Figure 11: Designing the installation view of *(solo)*

This work is also a modular one: multiple stainless steel cases which house more speaker arrays can be easily added or subtracted, and their location in relation to one another can be altered, be fitting of performance or large-scale installation. While *(solo)*'s primary purpose was intended for the creation of intimate listening spaces for one person, the expansion of the installation to two hemispheres with 12 ch audio and 48 speakers created a new and dynamic space for multiple listeners to engage with, as found in *(duo)* shown in Figure 3, above.

In *(duo)* the 48 speakers are housed behind the sphere, projecting through 4mm holes, making the audio signal resonate the metal casing. The sound qualities within the space as a result are less directional and reflective like *(solo)* but more resonant. The hemispherical cabinets can also be mounted in various configurations. In this iteration the sculptures were faced opposite one another.

Further documentation of *(duo)* may be viewed at:
https://www.youtube.com/watch?v=oKChuA_VAZI

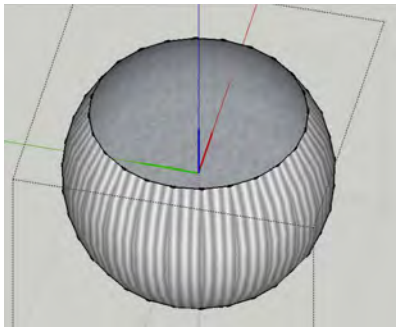


Figure 12: stainless steel casing design for *(solo)* and *(duo)*.

3.4. Software Design

In addition to a new hardware device, *(solo)* and *(duo)* make use of a customer software interface. The GUI will allow composers and performers greater expressive control while writing music and performing. The primary control of the GUI is to govern the volume of each section, and can easily be customised in terms of gain controls depending on the needs of the piece. It was written firstly with the idea for live performance, in order for the performer to have greater control over the numbers of signals coming from Ableton Live all at once.

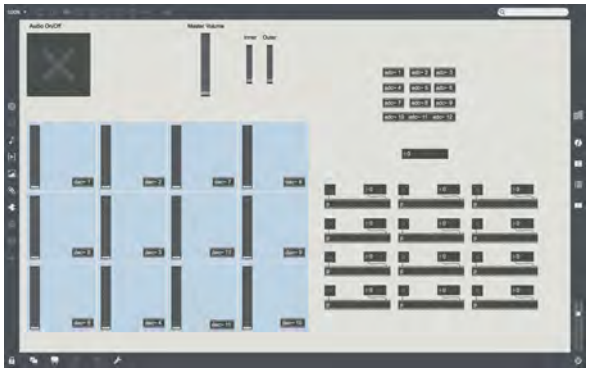


Figure 13: The *Speaker_Forest* diffusion interface: each channel is able to be controlled by a midi device and can be configured through a DAW for performance.

Figure 13 depicts the main GUI screen. It has an 'on/off' toggle to enable/disable audio output. Once toggled, it tells each **adc~** object to begin accepting audio from Soundflower and each **dac~** object to start outputting audio to the aggregate device. Soundflower is an external program which functions as a digital audio interface within the computer, allowing the user to route audio between programs (in this case Max and Ableton).

There are three main volume controls of Figure 13, seen at the top in the middle. There is a master gain, a gain for the inner channels (3, 12) and outer channels (1, 2, 7, 8, 9, 10, 11, 4, 5, 6). Each square in the blue grid corresponds to a speaker 'section' in the installation. The slider on each square allows for individual gain control of each of these speaker sections.

On the bottom right, there are twelve individual subpatchers which route each input signal to the 12 dac~ outputs, and allows output randomisation (detailed below). The toggle attached to each subpatcher enables/disables output randomisation, and the number sets the randomisation rate.

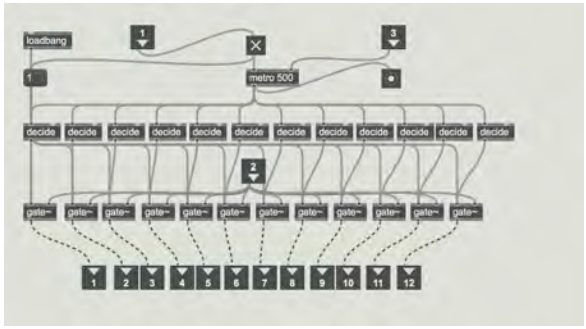


Figure 14: Max sub-patcher detailing

Figure 14 details the sub-patcher. It has 3 inputs (toggle randomisation, audio input, and randomisation rate) and 12 outputs (all audio), labelled 1 – 12 on the bottom. Each audio output receives its signal from **inlet 2** via a **gate~** object. Each **gate~** object can receive a message of either 0 (output disabled) or 1 (output enabled). When the Max patch is opened, the **loadbang** object ensures each outlet has output enabled at startup.

Inlet number 1 takes input from a toggle in the main patch, and controls the **metro** object, which outputs a

bang regularly at the specified rate (default 500ms). Inlet number 3 controls this rate. Every time the **metro** object bangs it triggers a **decide** object for each gate, which chooses randomly between 1 or 0, sending the result to the attached **gate~**. In this way, which gates (and therefore which outputs) are open or closed is randomised every time the metro bangs.

There is a randomise speaker function which will send an input signal from Ableton randomly between all 12 speakers to a certain rate, if toggled. This was to encourage new composers of spatial music the opportunity to interact with generative composition and to supply quickly gratifying spatial composition techniques to beginners.

The GUI allows composers and performers greater expressive control while writing and performing music. Like a conductor, the primary role of the GUI is to govern the volume of each section, and can easily be customised in terms of gain controls depending on the needs of the piece. It was written firstly with the idea for live performance in order for the performer to have greater control over the numbers of signals coming from Ableton all at once. This allows the composer greater flexibility and control writing spatial music as the GUI offers the opportunity for on the fly diffusion style music as well as fixed media. The opportunity to have multiple options to send the audio within the space proved crucial in the conceptualisation and composition of the audio.

4 . AUDIO & CONCEPT

Musical development was in part conceptual and in part informed by the installations themselves. It became apparent while writing music that the line between pre-composed musical ideas conceived prior to working with the rig versus ideas informed by composing with the installation were extremely blurred.

Speaker_Forest informed the musical structure through 'diffusion choreography'; inhales and exhales move inwardly and outwardly between the panels, repeated breathing forming a motif which became the base of the musical structure.

Building on Idhe's notion of the voice possessing an ‘auditory aura’ which ‘physically invades’ the body of the listener, the music itself was based on my first-person phenomenological experience in order to capture a 'memory space' [8]. In favour of using space as an alternative, Wishart's theory of the lattice [9] was put to the side; instead of just using diffusion 'spread' of the audio signal, this piece also engages location and panning techniques.

Conceptually, the music grew from the idea of sleeping next to my partner who suffers from serious breathing problems as there was a period of time where he was choking in his sleep without waking. This piece was written initially to explore the relationship we have to the body and technology but became a reflection of my relationship with my partner who suffers from asthma and chronic ill health. The piece forms my point of view as we are going to sleep, hearing regular breathing transform into irregular breath as I toss and turn. The music became an experiential mood-diary of what it felt like nursing someone you care about back to health. There are elements of trying to convince yourself that everything will be okay, combined with the panic of not knowing whether you're imagining sounds that are in your head or whether they're actually coming from your partner. This was explored through audio effects such as echo, panning and long reverb to symbolise the 'imagined' breath.

I realised that the piece as I was writing it became less about how we embody technology as humans, and more about how people are survived by technology and our reliance on it to keep us well. In my case, this was my dependence on my phone to call the ambulance in an emergency. The piece is about our relationship with technology as people, and the anxiety of never quite feeling comfortable with putting someone's health in the palms of a program.

In addition to this work, I approached friend and artist Mo Zareei to write a test piece for *Speaker_Forest.* As an experienced audio artist with Ableton Live and spatial composition, he quickly understood how to compose for the unusually large number of channels, exploring harsher, noise tones which bear resemblance to his brutalist sound sculptures[5]. It was quick and effective, the high pitches bouncing off the reverberant flat surfaces and creating a kind of 'twinkly blanket'. What I took away from Mo's contribution was how versatile the sound system is, and how it can be engaging for many different styles of music, be it ambient or otherwise. It was especially beneficial for me to see how our different compositional aesthetics both worked in such an immersive sound environment.

The audio component of *(solo)* was an iteration of the same concept explored in the *Speaker_Forest's* music, comprised only of breathing sounds. In this work, the inhaling begins at the centre-most point of the dome and exhales with rushed panning out to the edges of the dome. Whilst the audio track is a seven-minute loop, over time the breathing becomes quieter and quieter due to the installation being battery operated and running out of power. This part of the installation informed the piece with sense of mortality and finality. Breath emanating from a non-living object humanises it and embodies the traits we associate with living, making this sound experience very personal as it asks us to confront our mortality in a very intimate way.

In *(duo),* the sound is more obliquely related to themes of breath and the body via the desire to address the effect of urban and rural air pollution from anthropogenic sources upon natural, non-man-made environments. In turn, this pollution impacts our own bodies. I aimed to show that although despite the popular association of air pollution with urban and rural areas, air pollution produced by anthropogenic sources is impacting non-urban and non-agricultural areas too. Some of these areas are considered 'remote' from anthropogenic pollutant sources, showing that our impacts on humans are incredibly far-reaching. It also suggested that at many levels in where you live in the world, there are subtle changes in our own experiences of air pollution from taste, sight, smell, to hearing (not so much touch taste).

I decided to show the New Zealand bush which is often associated with our '100% pure NZ' brand, as slightly 'discoloured' with harsher sounds. Are they the sounds of cicadas or are you detecting in the background or is it something else, more unpleasant? The noise blends in with the natural environment making it harder to discern what is natural and what is not. Inspired by the ambient music of Alva Noto, I used tone generators to produce 15700 Hz - which is the same frequency value

5 http://m-h-z.net/soundsculptures/

as CRT TVs - as harsh sine waves and triangle waves. These tones circle the space, creating a tension over a five-minute period and are piercing in character, but subtle. Some people are unable to hear this high pitch at all, they will live happily listening to the bird song and think nothing of it. Some people with more acute hearing will recognise that this environment has been 'tampered with' by humans. This is very deliberate, so as to emphasise the permeation of human culture and industrial evolution upon the wider world. Based on my research of air pollution from the WHO, NASA, and New Zealand's Ministry of the Environment, I was able to make informed musical choices. Using musical elements against the backdrop of Wellington's Otari-Wilton bush walk engages with Brian Eno's concept of using ambient music as a 'tint' to emphasise aspects of the surrounding environment.

High frequencies such as 15700 Hz seems to have a range of responses from people due to the association with CRT TVs, and largely that tone is often felt in an unpleasant way. After watching this video[6] it was clear to me that as humans are made up largely of water, our bodies would respond in similar ways to sound waves, hence my compositional preference for high pitched sines over traditional noise music methodologies. The more complex the partial makeup, the more complex and possibly frenetic our responses could be.

The three installations were shown at 17 Tory St in Wellington, New Zealand during October 2016. Each work within same concrete room rectangular occupied an important role: hanging above audience members, hidden in the corner and two hemispheres facing one another, mounted within two plinths.

5. INSTALLATION AND PARTICIPANT EXPERIENCE

As the listener approaches the space they are greeted by intricate and complex – if not slightly intense - sounds from a somewhat indistinct source. As the configuration of *(duo)* and *Speaker_Forest* for installation is modular, this setup may be altered depending on the music written for it and how the composer would like the audience to interact with the sound for any given space. *(duo)* particularly draws on the relationship built between performer and reception, or in this case object and audience, inviting multiple listeners between the two pillars to engage.

Being that *(solo)* was battery powered, this directly impacted the nature of the audio. The audio work mimics breathing, of being up close right next to someone's mouth. Over the course of the piece, the breath would get quieter and quieter, and slowly 'dies'. It gives the work a feeling of mortality and finality which seldom comes with everlasting loop-based audio installation work.

(duo) took a different approach compositionally. A binaural field recording of the Wilton-Otari bush walk in Wellington was split between two hemispheres. The hemispheres faced one another, mounted in two plinths which helped to encourage people to sit and be immersed in the space between. In the audio, high-pitched tones mix with white noise to imply that the natural world is not totally untouched by human influence. It refers directly to the strong Wellington wind and how air pollution travels quickly over the city right into the bush. Although levels of air pollution are relatively low in New Zealand, the sonic allegory of the pollution, 15kHz frequency, generates many different reactions from listeners: indifference, pleasure or genuine pain/ discomfort.

The immersive nature of all three of these installations, coupled with the breathing motifs in *Speaker_Forest* and *(solo)* encourage the listener to embody the human qualities of the audio and 'lose themselves' in a miniature sound world. In the context of *(duo)*, the intense feeling some people get from the 15kHz frequency explores the barrier between the natural world and the man-made one.

6. FUTURE WORK/ CONCLUSION

The research and development of these smaller sound installations are necessary and important in a few simple ways; they are typically more cost-effective, portable and are modular in essence to suit a variety of different applications. It shows that this form of spatial sound artwork can be a more sustainable stream of creating artwork in the future.

Further development along the lines of *(solo)* and *(duo)* would be to expand the scope of the project itself. Adding extra domes, amps and speakers is a clear avenue forward Another option would be to try experimenting with the size and shape of the domes to test reverberant and resonant qualities of the space and metal. Improvements can be made to the current prototypes as well; polishing each dome until it has a mirror finish.

Speaker_Forest has the potential to grow into a large-scale work. Given that the premise of the work is modular and designed for more panels to be added, the piece could expand the number of aggregate audio devices and audio channels. The sculptural placement of the panels could likewise be further explored. This could include but is not limited to adding more speakers, reconfiguring their placement on each installation or have them hanging them at different heights to acoustically activate different parts of a room.

Experimenting with speakers of larger sizes could open up further compositional possibilities, as the current frequency response of the smaller speakers sounds somewhat tinny. My current idea would be to make a speaker array of larger speakers, housed in cinderblocks which line the base of a room. Above that would hang four to eight panels of the *Speaker_Forest*. Once such application could be for small performances with live electronics.

Another opportunity for development could be through the GUI: with an increase in scope, location and placement of the panels could open up new ideas for generative composition which has been briefly explored within the Max patch. This could include, but is not limited to, developing extra functions which toggle a set number of speakers of their choice, rather than a default to all speakers. It suggests that in tandem with elements of pitch, timbre and time, the role of space in musical composition has been largely overlooked. In order to develop our understanding of sound environments and

6 'Star-Shaped Droplet Levitated By Sound', YouTube, accessed Tuesday 12th April, https://www.youtube.com/watch?v=MKMf7PCkoZQ

their impacts on musical composition, we should explore interactive spatial sound choreography further.

Ultimately I think that the creation and development of spatialised audio systems should stay as modular as possible to fit the composer's needs for multiple applications. The brief piece by Zareei demonstrated to me how keeping the installation open-ended really opens up new possibilities for immersive and intimate sound spaces that play against the architecture of any given space. I think in these three installations there is promising new ground to explore the space between intimate listening environments which cater to few, or to many. Although the musical works did not directly address the need for a 'safer space', I feel as though they successfully engaged intimate and embodied performances of my own phenomenological breath. As far as using spatial sound installation as an empathic tool, I believe that the immersive nature of the work lend themselves to more embodied experiences which are unique to the individual who engages with the work.

7. REFERENCES

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